Code No: 158AU **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year II Semester Examinations, September - 2022
DISTRIBUTED SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the trends in distributed systems.
b) Explain the architectural model of distributed system. [7+8]

2.a) Elaborate the issues for Design of distributed systems.
b) What are the characteristics of interprocess communication? [7+8]

3.a) Explain the functions provided by operating system.
b) Explain Invocations between address spaces. [7+8]

4. With a neat diagram, explain file service architecture. [15]

5.a) Explain the differences between IP and overlay routing for peer-to-peer applications.
b) Explain in detail about Routing Overlays. [8+7]

6.a) Explain the Cristian's method for synchronizing clocks.
b) Explain distributed mutual exclusion. [8+7]

7.a) What is a lock? Explain the locking rules for nested transactions.
b) Describe the operations for two-phase commit protocol. [8+7]

8.a) Explain the architectural model for the management of replicated data.
b) Discuss whether message passing or DSM is preferable for fault-tolerant applications. [7+8]

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